

# Synthesis and Application of Chiral $\beta$ -Amino Disulfides as Ligands for the Enantioselective Addition of Diethylzinc to Aldehydes

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**ABSTRACT** A new class of chiral  $\beta$ -amino disulfides was synthesized from readily available and inexpensive starting materials by a straightforward method and their abilities as ligands were examined in the enantioselective addition of diethylzinc to aldehydes. Enantiomeric excesses of up to 99% have been obtained using 0.5 mol % of the chiral catalysts. *Chirality* 20:839–845, 2008. © 2008 Wiley-Liss, Inc.

**KEY WORDS:** asymmetric catalysis; chiral ligands; thiols; sulfur; zinc

## INTRODUCTION

Addition of organometallic reagents to carbonyl compounds is one of the most common and fundamental reactions for the formation of carbon–carbon bonds, and the asymmetric version of this reaction is particularly useful for generating optically active secondary alcohols.<sup>1–3</sup> The importance of this structural feature arises mainly from the fact that it is part of many natural products or precursors as an important synthetic intermediate for various other functionalities, e.g., amine, halide, ester, and ether. Concerning this subject, several highly stereoselective additions to prochiral aldehydes have been reported,<sup>4–8</sup> and among the various organometallic compounds,<sup>9,10</sup> di-organozincs serve as excellent alkyl nucleophiles.<sup>11–17</sup> The development and functional understanding of chiral ligands for catalytic asymmetric synthesis is of increasing importance in modern synthetic chemistry; and in particular, the addition of diethylzinc to aldehydes has become a classical test to design new ligands for asymmetric catalysis.<sup>1</sup> Chiral  $\beta$ -amino alcohols are often used as ligands in this reaction<sup>18–24</sup> because of its easy access normally direct from the respective amino acids. Rationalizations about the stereoselectivity in asymmetric addition of dialkylzinc reagents to aldehydes using these ligands as model were performed by various research groups, giving an important tool to understand mechanistically the way and to improve the development of new classes of chiral catalysts.<sup>25</sup>

In the last few years, a significant improvement in enantioselectivity has been observed by the use of amino sulfur or selenium moieties as catalysts for the enantioselective addition of diethylzinc to aldehydes.<sup>26–32</sup> Sulfur containing ligands<sup>33</sup> like  $\beta$ -amino thiols and  $\beta$ -amino disulfides derivatives from ephedrine,<sup>34–36</sup> norephedrine,<sup>37–39</sup> (S)-proline,<sup>40–42</sup> (S)-phenylglycine,<sup>43</sup> (S)-valine,<sup>44,45</sup> (R)-cysteine,<sup>5,46–50</sup> (R)-cystine,<sup>51</sup> and others<sup>52–58</sup> have gained more and more interest as catalysts in this stereoselective reac-

tion. In a similar way, chiral amino diselenides have been efficiently used as catalysts for this addition,<sup>28–32</sup> but due to the most difficult accessibility of these compounds, only a few number of reports can be found in literature.

As part of our broader program to explore the preparation and use of chiral organochalcogen compounds in asymmetric catalysis,<sup>59–64</sup> especially ligands binding via sulfur or selenium, we describe in this article the development of a new class of [S, N]-ligands in a straightforward synthetic route, using inexpensive and easily available starting materials and their application as chiral ligands in the enantioselective addition of diethylzinc to aldehydes.

## EXPERIMENTAL SECTION

### General Procedures

Melting Points are uncorrected. Optical rotations were measured on a Perkin-Elmer 341 Polarimeter. <sup>1</sup>H and <sup>13</sup>C NMR spectra were recorded at 400 and 100 MHz, respectively, with tetramethylsilane as internal standard. High-resolution mass spectra were recorded on a Bruker BioApex 70 eV spectrometer. Gas chromatography (GC) was performed using a Varian 3800 gas chromatograph with Hydrodex  $\beta$ -3P column. HPLC analyses were carried out on a Shimadzu SCL-10 Avp chromatograph using a Diacel Chiralcel OD column; solvent, 99:1 hexane/isopropanol; flow rate 0.5 mL/min; 254 nm detection. Column chromatography was performed using Merck Silica Gel (230–400 mesh) following the methods described by Still et al.<sup>65</sup> Thin layer chromatography (TLC) was performed using

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Merck Silica Gel GF<sub>254</sub>, 0.25 mm thickness. For visualization, TLC plates were either placed under ultraviolet light, or stained with iodine vapor, or acidic vanillin. Most reactions were monitored by TLC for disappearance of starting material. The following solvents were dried and purified by distillation from the reagents indicated: tetrahydrofuran from sodium with a benzophenone ketyl indicator; dichloromethane from calcium hydride; acetonitrile from phosphorus pentoxide. All other solvents were ACS or HPLC grade unless otherwise noted. Air- and moisture-sensitive reactions were conducted in flame-dried or oven dried glassware equipped with tightly fitted rubber septa and under a positive atmosphere of dry argon. Reagents and solvents were handled using standard syringe techniques. Temperatures above room temperature were maintained by use of a mineral oil bath with an electrically heated coil connected to a Variac controller.

**General procedure for the synthesis of amino disulfides 4.** In a 50-mL round-bottomed flask with a reflux condenser, under an argon atmosphere, lithium disulfide was generated by reaction of elemental sulfur (0.0384 g, 1.2 mmol) with lithium triethylborohydride (1 M in THF, 1.2 mL, 1.2 mmol) in dry THF (5 mL). The suspension was heated at reflux and stirred for at least 20 min, and a THF (10 mL) solution of aziridine **2** (1.0 mmol) was added drop wise within 20 min. The resulting solution was stirred for 12 h at room temperature. The mixture was quenched with a saturated NH<sub>4</sub>Cl solution (20 mL), extracted with CH<sub>2</sub>Cl<sub>2</sub> and the combined organic fractions were collected, dried over MgSO<sub>4</sub> and filtered; the solvent was then removed in vacuo yielding crude products **4a–d**, which were purified by flash chromatography (Hexane:ethyl acetate 90:10).

**$\beta$ -Amino disulfide 4a.** Yield: 0.447 g (84%). White solid. m.p. 144–146°C.  $[\alpha]_D^{20} = -11$  ( $c = 1.0$ , CH<sub>2</sub>Cl<sub>2</sub>). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 7.30$ – $7.18$  (m, 10 H), 4.94–4.92 (m, 2 H), 4.12–4.09 (m, 2 H), 2.91–2.81 (m, 8 H), 1.42 (s, 18 H) ppm. <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>):  $\delta = 155.2$ , 137.5, 129.3, 128.4, 126.5, 79.4, 51.1, 43.4, 38.8, 28.3 ppm. HRMS (ESI): calcd for C<sub>28</sub>H<sub>40</sub>N<sub>2</sub>NaO<sub>4</sub>S<sub>2</sub> [M + Na]<sup>+</sup> 555.232722; found 555.232721. The enantiomeric purity was determined by HPLC analysis (column, Chiralcel-OD; eluent, hexane/2-propanol 98:02; flow rate, 1 mL/min; *R* isomer,  $t_R = 16.3$  min, *S* isomer,  $t_R = 17.2$  min) and found to be >99.9%.

**$\beta$ -Amino disulfide 4b.** Yield: 0.345 g (79%). White solid. m.p. 104–105°C.  $[\alpha]_D^{20} = +69$  ( $c = 1.0$ , CH<sub>2</sub>Cl<sub>2</sub>). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 4.78$ – $4.76$  (m, 2 H), 3.69–3.67 (m, 2 H), 2.90–2.89 (m, 4 H), 1.94–1.90 (m, 2 H), 1.44 (s, 18 H), 0.96 (d,  $J = 8.2$  Hz, 6 H), 0.92 (d,  $J = 8.2$  Hz, 6 H) ppm. <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>):  $\delta = 155.8$ , 79.2, 55.4, 42.8, 32.5, 30.8, 19.5, 17.7 ppm. HRMS (ESI): calcd for C<sub>20</sub>H<sub>40</sub>N<sub>2</sub>NaO<sub>4</sub>S<sub>2</sub> [M + Na]<sup>+</sup> 459.232722; found 459.232721.

**$\beta$ -Amino disulfide 4c.** Yield: 0.362 g (78%). White solid. m.p. 67–69°C.  $[\alpha]_D^{20} = -37$  ( $c = 1.0$ , CH<sub>2</sub>Cl<sub>2</sub>). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 4.95$ – $4.91$  (m, 2 H), 3.91–3.87 (m, 2 H), 2.82–2.81 (m, 2 H), 2.79–2.78 (m, 2 H),

1.70–1.67 (m, 2 H), 1.44 (s, 22 H), 0.94 (d,  $J = 9.4$  Hz, 12 H) ppm. <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>):  $\delta = 155.4$ , 79.1, 48.3, 45.2, 42.9, 28.4, 25.0, 23.1, 22.0 ppm. HRMS (ESI): calcd for C<sub>22</sub>H<sub>44</sub>N<sub>2</sub>NaO<sub>4</sub>S<sub>2</sub> [M + Na]<sup>+</sup> 487.264022; found 487.265880.

**$\beta$ -Amino disulfide 4d.** Yield: 0.371 g (80%). White solid. m.p. 70–72°C.  $[\alpha]_D^{20} = +10$  ( $c = 1.0$ , CH<sub>2</sub>Cl<sub>2</sub>). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 4.79$ – $4.76$  (m, 2 H), 3.75–3.72 (m, 2 H), 2.99–2.89 (m, 4 H), 1.66–1.64 (m, 2 H), 1.45 (s, 18 H), 1.12–1.10 (m, 4 H), 0.93–0.89 (m, 12 H) ppm. <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>):  $\delta = 155.6$ , 79.1, 54.3, 42.3, 37.2, 28.4, 24.9, 15.4, 11.5 ppm. HRMS (ESI): calcd for C<sub>22</sub>H<sub>44</sub>N<sub>2</sub>NaO<sub>4</sub>S<sub>2</sub> [M + Na]<sup>+</sup> 487.264022; found 487.265879.

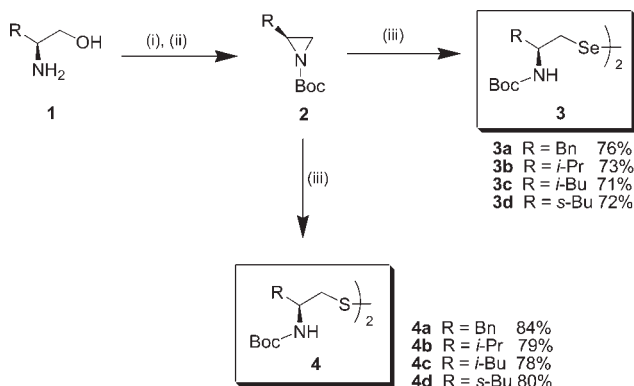
**Procedure for the synthesis of thioether 6.** In a 25-mL round-bottomed flask, under argon atmosphere, the thiolate anion was generated by the reaction of diethyl disulfide (0.0734 g, 0.6 mmol) with NaBH<sub>4</sub> (0.057 g, 1.5 mmol) in a mixture of THF (1.5 mL) and EtOH (0.5 mL). The suspension was allowed to stir for at least 20 min, and a THF (10 mL) solution of aziridine **2a** (0.233 g, 1.0 mmol) was added drop wise within 20 min. The resulting solution was stirred for 12 h at room temperature. The mixture was quenched with a saturated NH<sub>4</sub>Cl solution (20 mL), extracted with CH<sub>2</sub>Cl<sub>2</sub> and the combined organic fractions were collected, dried over MgSO<sub>4</sub> and filtered; the solvent was then removed in vacuo yielding crude thioether **6**, which was purified by flash chromatography (Hexane:ethyl acetate, 90:10). Yield: 0.220 g (75%) Spectral data consistent with those reported in the literature.<sup>66</sup>

**General procedure for the enantioselective addition of diethylzinc to aldehydes.** In a 25-mL round-bottomed flask, a solution of toluene (7 mL), aldehyde (1.0 mmol), and the ligand **4a–d** (0.5 mol%, 12.5  $\mu$ mol) was stirred for 30 min at room temperature. Diethylzinc (1 M in hexane, 2.5 mL, 2.5 mmol) was slowly injected and stirring was continued for 24 h at room temperature. Cooling (0°C) of the reaction mixture was followed by the slow addition of HCl (aq, 1 M, 5 mL). The organic layer was separated and washed again with HCl (aq, 1 M, 3 mL). Drying with MgSO<sub>4</sub>, filtration, and evaporation of the solvent in vacuo yielded the crude secondary alcohols. Purification was performed by bulb-to-bulb distillation under reduced pressure (ca. 0.1 mbar).

**(*R*)-1-phenylpropan-1-ol.** Yield: 0.122 g (90%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 7.32$ – $7.13$  (m, 5 H), 4.54 (t,  $J = 6.6$  Hz, 1 H), 1.87–1.67 (m, 2 H), 0.85 (t,  $J = 7.4$  Hz, 3 H) ppm. <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>):  $\delta = 144.5$ , 128.3, 127.3, 125.9, 75.9, 31.8, 10.0 ppm.

**(*R*)-1-(2-methoxyphenyl)propan-1-ol.** Yield: 0.133 g (80%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 7.30$ – $7.13$  (m, 2 H), 6.69–6.67 (m, 2 H), 4.77 (t,  $J = 6.4$  Hz, 1 H), 3.75 (s, 3 H), 2.94 (m, 1 H), 1.75 (q,  $J = 7$  Hz, 2 H), 0.91 (t,  $J = 7.4$  Hz, 3 H) ppm. <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>):  $\delta = 156.2$ , 132.3, 127.8, 126.7, 120.4, 110.2, 71.5, 30.0, 10.2 ppm.

**(*R*)-1-(4-methoxyphenyl)propan-1-ol.** Yield: 0.125 g (75%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta = 7.29$ – $7.01$  (m, 4



**Scheme 1.** Synthesis of  $\beta$ -amino diselenides **3** and  $\beta$ -amino disulfides **4** reagents and conditions: (I)  $\text{Boc}_2\text{O}$ ,  $\text{CH}_3\text{CN}$ , r.t., 3 h; (II) KOH, TSCl, THF, reflux, 4 h; (III)  $\text{Li}_2\text{Se}_2$  or  $\text{Li}_2\text{S}_2$ , THF, r.t., 12 h.

H), 4.35 (t,  $J = 6.1$  Hz, 1 H), 3.55 (s, 3 H), 1.85–1.63 (m, 2 H), 0.91 (t,  $J = 7.0$  Hz, 3 H) ppm.  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ ):  $\delta = 158.4, 137.9, 128.9, 113.4, 75.3, 55.1, 31.9, 10.0$  ppm.

**(*R*)-1-(pyridin-2-yl)propan-1-ol.** Yield: 0.107 g (78%).  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta = 8.47$  (m, 1 H), 7.57–7.01 (m, 3 H), 4.65 (t,  $J = 5.7$  Hz, 1 H), 1.93–1.75 (m, 2 H), 1.12–0.87 (m, 3 H) ppm.  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ ):  $\delta = 162.1, 150.1, 137.0, 122.6, 122.1, 79.1, 31.6, 10.0$  ppm.

**(*R*)-dodecan-3-ol.** Yield: 0.100 g (54%).  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta = 3.35$ – $3.29$  (m, 1 H), 1.48–1.19 (m, 19 H), 0.91–0.86 (m, 6 H) ppm.  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ ):  $\delta = 70.5, 36.8, 31.9, 31.5, 30.1, 29.6, 29.5, 29.4, 25.2, 22.7, 14.2, 10.2$  ppm.

**(*R*)-octan-3-ol.** Yield: 0.066 g (51%).  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta = 3.65$ – $3.51$  (m, 1 H), 1.59–1.23 (m, 10 H), 1.09–0.86 (m, 6 H) ppm.  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ ):  $\delta = 73.2, 36.8, 31.9, 30.0, 25.3, 22.6, 13.9, 9.8$  ppm.

#### Conditions for the Analysis of Chiral Secondary Alcohols

**Chiral capillary GC.** Hydrodex  $\beta$ -3P column 25 m  $\times$  0.25 mm. Carrier gas,  $\text{H}_2$  (5 mL/min); detector, FID, 280°C; injector, 220°C.

**Chiral HPLC.** Chiralcel OD column; solvent, 99:1 hexane/isopropanol; flow rate 0.5 mL/min; 254 nm UV detector.

The racemic alcohols products were obtained by addition of  $\text{EtMgBr}$  to aldehydes. The conditions of analysis and retention times of the *R* and *S* isomers have been reported elsewhere.<sup>67,68</sup>

## RESULTS AND DISCUSSION

We have previously reported that the use of only 0.5 mol % of the chiral amino diselenides **3a–d** as ligands in the enantioselective addition of diethylzinc to aldehydes, afforded the respective secondary alcohols in up to 99% *ee*.<sup>69</sup> To the best of our knowledge, these ligands were the first class of chiral amino diselenides with the selenium

atom attached to an alkyl group. From now, we set out to extend this study through the synthesis of other similar ligands, possessing potentially stereogenic chalcogenide donors. Scheme 1 depicts the synthesis of these two classes of ligands:  $\beta$ -amino diselenides **3a–d**, previously reported, and the new  $\beta$ -amino disulfides **4**. These compounds would allow us to study electronically and structurally the effects of these different catalysts and hopefully provide insight into further ligand design.

The chiral amino chalcogenides **3a–d** and **4a–d** were easily prepared from the corresponding commercially available  $\alpha$ -amino alcohols **1**, which were further quantitatively converted into the *N*-Boc-protected derivatives by reaction with di-*tert*-butyl dicarbonate in acetonitrile. Chiral aziridines **2** were obtained in good yields by treatment of *N*-Boc amino alcohols with *p*-toluenesulfonyl chloride and potassium hydroxide in boiling THF. Finally, the sulfur or selenium atom was efficiently introduced by regioselective ring opening employing  $\text{Li}_2\text{Y}_2$ <sup>70</sup> ( $\text{Y} = \text{S}, \text{Se}$ ) as nucleophile. The attack was selective in the less hindered carbon<sup>71,72</sup> of aziridines ring **2a–d**, furnishing the aliphatic chiral amino diselenides **3a–d** and disulfides **4a–d** without any loss of enantiomeric purity, as determined by HPLC analysis using Daicel Chiralcel OD column. The regiochemistry was determined by analysis of NMR data, confirming the regioselective aziridine ring opening without detection of a second regioisomer.

With this sterically and electronically varied set of enantiopure dichalcogenides, we have examined their efficiency as chiral ligands in the enantioselective addition of diethylzinc to aldehydes. Aiming to determine the optimum conditions, we performed studies about effects that can influence this catalysis, such as load of catalyst and temperature.

**TABLE 1.** Enantioselective addition of diethylzinc to benzaldehyde varying the loading of ligand **4a**

| Entry          | Loading of <b>4a</b> (mol %) | Yield <sup>a</sup> (%) | <i>ee</i> <sup>b</sup> (%) |
|----------------|------------------------------|------------------------|----------------------------|
| 1              | 10                           | 92                     | 95 (R)                     |
| 2              | 5                            | 92                     | 95 (R)                     |
| 3              | 2.5                          | 91                     | 95 (R)                     |
| 4              | 1                            | 91                     | 95 (R)                     |
| 5              | 0.5                          | 90                     | 95 (R)                     |
| 6              | 0.25                         | 43                     | 62 (R)                     |
| 7 <sup>c</sup> | 0.5                          | 94                     | 83 (R)                     |
| 8 <sup>d</sup> | 0.5                          | 48                     | 90 (R)                     |

<sup>a</sup>Determined by GC analysis.

<sup>b</sup>Determined by chiral GC analysis with use of a Hydrodex  $\beta$ -3P column; the absolute configuration was determined by comparing the sign of the optical rotation.

<sup>c</sup>Reaction was carried out at 50°C for 20 h.

<sup>d</sup>Reaction was carried out at 0°C for 48 h.

In a preliminary stage, ligand **4a** was tested in the addition of diethylzinc to benzaldehyde to determine the optimal loading of catalyst. Yields and enantioselectivities for 1-phenyl-1-propanol have been collected in Table 1.

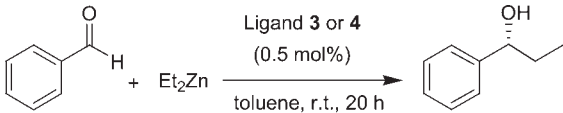
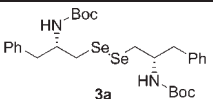
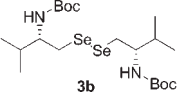
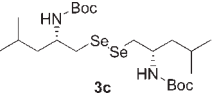
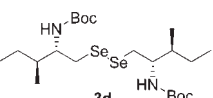
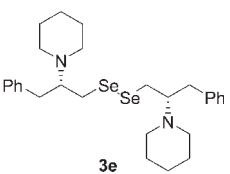
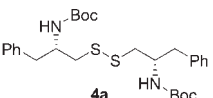
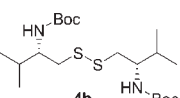
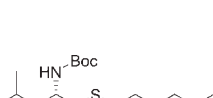
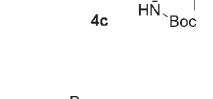
Concentrations varying from 0.25 to 10 mol % were tested and it could be observed that the *ee* of the alkylated product remained very high in the most of the cases (Table 1, entries 1–6). Even when 0.5 mol % of ligand **4a** was used, enantiomeric excess of 1-phenyl-1-propanol was 95% and the conversion was still high (Table 1, entry 5). However, at lower catalyst concentration, yield and enantiomeric excess of this secondary alcohol markedly decreased (Table 1, entry 6). This result can be rationalized due to the high affinity of sulfur with zinc, forming a highly effective catalytic cycle. At this point, it has to be mentioned that the described ligand systems display a high reactivity compared to most of the other ligand systems described in the literature.<sup>73,74</sup>

In addition, we investigated the temperature effect on the enantiomeric excess of the product in the same (1,2)-addition reaction. We also observed that this process is sensible to this effect, in terms of yield and selectivity. When the reaction was carried out at room temperature, the conversion was of 90%, and the *ee* of 95% (Table 1, entry 5). By increasing the temperature, a negative effect in terms of *ee* was observed while the yields remained practically constant (Table 1, entry 7). Probably, the increment of temperature supply to this reaction and additional energy necessary to form a nonfavorable transition state, decreasing the *ee* in this case. By lowering the temperature to 0°C, we observed a drastic decrease in the yield (Table 1, entry 8), probably due to a decreasing in the efficiency of complexation; while the *ee* remained practically the same obtained at room temperature.

With these results in hand, the efficiency of the other disulfides prepared was also examined. The reactions were performed under standard conditions: 0.5 mol % of ligand **4**, benzaldehyde (1 mmol), diethylzinc (2.5 mmol) in toluene, at room temperature. Aiming to perform a comparative study between ligands **3** and **4** to verify the influence of chalcogen atom as effective ligand, the enantioselectivities observed with diselenides **3**<sup>69</sup> to the respective secondary alcohols are also given in the following Tables.

As shown in Table 2, all the ligands has presented high levels of enantiocontrol, affording (*R*)-1-phenyl-1-propanol in enantiomeric excesses ranging from 91 to 95%. The steric hindrance from the different alkyl groups in the side chains of aminoalcohols plays an important role in the selectivity, and the ligands **3a** and **4a** (Table 2, Entries 1 and 6) shown better levels of enantioselection in comparison with the other less bulky *R*-substituents. Concerning the influence of the chalcogen atom, selenium and sulfur showed practically the same efficiency. This result can be rationalized by the high tendency of both atoms coordinate to a metal center (zinc) forming an efficient catalytic complex. The steric effect on the amine moiety was also studied using the modified catalyst **3e**.<sup>69</sup> Although this ligand had shown good performance, the results obtained with **3a** are still higher in terms of conversion and enantio-

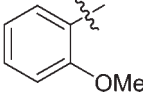
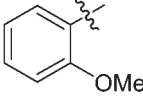
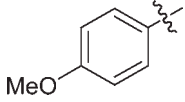
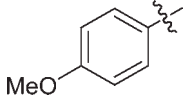
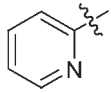
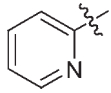
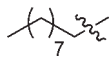
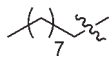
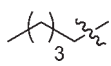
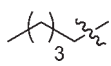
**TABLE 2. Screening of ligands 3a–e and 4a–d in the catalytic enantioselective addition of diethylzinc to benzaldehyde**

|  |                                                                                      |                        |                            |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------|----------------------------|
| Entry                                                                              | Ligand                                                                               | Yield <sup>a</sup> (%) | <i>ee</i> <sup>b</sup> (%) |
| 1                                                                                  |    | 91                     | 95 (R)                     |
| 2                                                                                  |    | 71                     | 91 (R)                     |
| 3                                                                                  |    | 80                     | 92 (R)                     |
| 4                                                                                  |    | 72                     | 91 (R)                     |
| 5                                                                                  |  | 82                     | 90 (R)                     |
| 6                                                                                  |  | 90                     | 95 (R)                     |
| 7                                                                                  |  | 70                     | 92 (R)                     |
| 8                                                                                  |  | 79                     | 94 (R)                     |
| 9                                                                                  |  | 70                     | 92 (R)                     |

<sup>a</sup>Determined by GC analysis.

<sup>b</sup>Determined by chiral GC analysis using a Hydrodex β-3P column; and the absolute configuration was determined comparing the sign of optical rotation.

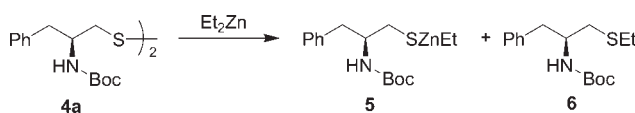
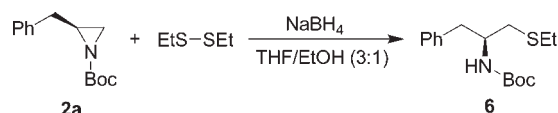
**TABLE 3.** Enantioselective addition of diethylzinc to aldehydes using **3a** or **4a**

| Entry           | Ligand    | R <sup>1</sup>                                                                      | Yield <sup>a</sup> (%) | ee (%)  |
|-----------------|-----------|-------------------------------------------------------------------------------------|------------------------|---------|
| 1 <sup>a</sup>  | <b>3a</b> |    | 93                     | 95 (R)  |
| 2 <sup>b</sup>  | <b>4a</b> |    | 80                     | 95 (R)  |
| 3 <sup>b</sup>  | <b>3a</b> |    | 93                     | >99 (R) |
| 4 <sup>b</sup>  | <b>4a</b> |    | 75                     | >99 (R) |
| 5 <sup>c</sup>  | <b>3a</b> |    | 85                     | 91 (R)  |
| 6 <sup>c</sup>  | <b>4a</b> |    | 78                     | 98 (R)  |
| 7 <sup>c</sup>  | <b>3a</b> |   | 56                     | 45 (R)  |
| 8 <sup>c</sup>  | <b>4a</b> |   | 54                     | >99 (R) |
| 9 <sup>c</sup>  | <b>3a</b> |  | 63                     | >99 (R) |
| 10 <sup>c</sup> | <b>4a</b> |  | 51                     | 70 (R)  |

<sup>a</sup>Conversion was determined by GC analysis.<sup>b</sup>Determined by chiral GC analysis using a Hydrodex  $\beta$ -3P column; the absolute configuration was determined by comparing the sign of optical rotation.<sup>c</sup>Determined by chiral HPLC analysis using a Daicel Chiralcel OD column.

meric excess, as well as the simplicity of the catalyst (Table 2, compare entries 1 vs. 5).

To extend the scope of the optimal ligand **4a** as chiral catalysts in this reaction, several aromatic and aliphatic aldehydes were screened. As shown in the Table 3, for aromatic aldehydes, high yields and levels of enantioselection of the respective (*R*)-secondary alcohols were observed by the use of both catalysts (entries 1–6) and in the best cases, *ees* up to 99% were achieved (see entries 3 and 4). When the less reactive aliphatic aldehydes were employed, the results had shown an unexpected way. By using the diselenide **3a**, the selectivity was excellent for hexanal (minor chain), and very poor to decanal (Table 3,

**Scheme 2.** Formation of the catalytically active species.**Scheme 3.** Synthesis of thioether **6**.

entries 7 and 9). Surprisingly, by using the disulfide **4a** the selectivity observed had an opposite way: (*R*)-dodecan-3-ol was obtained in up to 99% *ee* while (*R*)-octan-3-ol was achieved in only a moderated enantiomeric excess (Table 3, entries 8 vs. 10). Apparently, the size of the chalcogen atom in the structure of the ligand plays an important influence in terms of selectivity.

The stereochemistry of the products is in accordance with the mechanistic rationalization described in Noyori's work.<sup>1,75,76</sup>

To elucidate the mechanistic aspects, we assumed that the disulfide linkage is cleaved by nucleophilic attack of diethylzinc, resulting in the thiolate **5** and the thioether **6** (Scheme 2), according to the results reported by Kellogg<sup>35</sup> and Wirth.<sup>30</sup>

Aiming to identify the catalytically active species, we prepared the compound **6** via **2a** ring-opening reaction by attack of EtSSEt/NaBH<sub>4</sub><sup>77</sup> as shown in Scheme 3.

We believe that the active catalyst of the reaction is the species **5**, since in an additional experiment, the thioether **6** did not catalyze the alkylation reaction of benzaldehyde.

## CONCLUSIONS

In summary, a new series of chiral  $\beta$ -amino disulfides was synthesized in a straightforward synthetic route from readily available and inexpensive starting materials. These compounds have been evaluated in the enantioselective addition of diethylzinc to aldehydes, furnishing the corresponding optically active secondary alcohols in high yields and enantiomeric excesses by using a very small amount of catalyst. Most importantly, by the use of these compounds it was possible to perform a comparative study with the analog diselenides **3** reported by our group. As a result of this study, a larger class of complementary dichalcogenide ligands **3** and **4** for the addition of diethylzinc to aldehydes, especially to aliphatic aldehydes, is now available. Studies dealing with the mechanism of the reaction and application of this catalyst system in other asymmetric catalytic reactions<sup>78</sup> and as building blocks for peptides synthesis<sup>79</sup> are currently in progress in our laboratory.

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